```
function [a,b,c1,c2,emp_vario] = ps_fit_vario(x,y,z,model,a0,b0,c10,c20)

% Estimation of variogram model
%
% Input:    - x                 x-coordinates
%           - y                 y-coordinates
%           - z                 z-values
%           - model             variogram model:
%                               2 = exponential
%                               3 = gaussian
%                               4 = spherical
%           - a0                initial value range
%           - b0                initial value hole effect
%           - c10               initial value sill
%           - c20               initial value nugget
%
% Output:   - a                 estimated range
%           - b                 estimated hole effect
%           - c1                estimated sill
%           - c2                estimated nugget
%           - emp_vario         empirical variogram
%
% ------------------------------------------------------------------------
% File............: ps_fit_vario.m
% Version & Date..: 1.7.2.16, 12-DEC-2009
% Authors.........: Freek van Leijen
%                   Delft Institute of Earth Observation and Space Systems
%                   Delft University of Technology
% ------------------------------------------------------------------------
%
% This software is developed by Delft University of Technology and is
% intended for scientific use only. Applications for commercial use are
% prohibited.
%
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%
% Change log
%


% -----------------------------------------------------
% Initialize
% -----------------------------------------------------

global fig detail_plots visible_plots


% -----------------------------------------------------
% Calculate raw variogram
% -----------------------------------------------------

Nx = size(x,1);
xa = repmat(int16(x),1,Nx);
xb = xa';
ya = repmat(int16(y),1,Nx);
yb = ya';
h1 = repmat(single(NaN),size(xa));
Nbuffer = ceil(Nx/1000);
for v = 1:Nbuffer
  sl = (v-1)*1000+1;
  el = min(v*1000,Nx);
  for w = 1:Nbuffer
    sp = (w-1)*1000+1;
    ep = min(w*1000,Nx);
```

```
    h1(sl:el,sp:el) = sqrt((xa(sl:el,sp:ep)-xb(sl:el,sp:ep)).^2+...
                     (ya(sl:el,sp:ep)-yb(sl:el,sp:ep)).^2);
  end
end
clear xa xb ya yb

za = repmat(single(z),1,Nx);
h2 = 0.5*((za-za').^2);
clear za

h1 = h1(:);
h2 = h2(:);
h1(1:Nx+1:Nx*Nx) = []; %remove h1==0
h2(1:Nx+1:Nx*Nx) = [];

if strcmp(detail_plots,'y')
  fig = fig+1;
  figure(fig);hold on
  if strcmp(visible_plots,'n')
    set(gcf,'visible','off');
  end
  plot(h1,h2,'*');
  xlabel('Distance [m]')
  ylabel('Variogram [rad^2]')
end


% ------------------------------------------------------
% Calculate experimental variogram
% ------------------------------------------------------

max_dist = max(h1(:));
lags = [0:max_dist/50:max_dist];
Nlags = length(lags)-1;
emp_vario = repmat(NaN,Nlags,3);
for v = 1:Nlags
    index = find((h1>=lags(v))&(h1<lags(v+1)));
    emp_vario(v,1) = length(index);
    if emp_vario(v,1)~=0
      emp_vario(v,2) = mean(h2(index));
    end
    emp_vario(v,3) = (lags(v)+lags(v+1))/2;
end
index = find(isnan(emp_vario(:,2)));
emp_vario(index,:) = [];

if strcmp(detail_plots,'y')
  figure(fig);
  plot(emp_vario(:,3),emp_vario(:,2),'r*','markersize',10)
end

emp_vario = emp_vario(1:round(Nlags/2),:);
Nlags = size(emp_vario,1);


% ------------------------------------------------------
% Estimate experimental variogram
% ------------------------------------------------------

lam = [a0 c10 c20];

switch model
  case 1
    error(['You can not estimate the nugget independently, choose ' ...
```

```
           'another model (nugget included)']);
  case 2 %exponential
    [lam,resnorm,residual,exitflag] =
lsqcurvefit(@ps_variogram_exp,lam,emp_vario(:,3),emp_vario(:,2),[0 0 0],
[],optimset('Display','off'));
  case 3 %gaussian
    [lam,resnorm,residual,exitflag] =
lsqcurvefit(@ps_variogram_gaus,lam,emp_vario(:,3),emp_vario(:,2),[0 0 0],
[],optimset('Display','off'));
  case 4 %spherical
    [lam,resnorm,residual,exitflag] =
lsqcurvefit(@ps_variogram_spher,lam,emp_vario(:,3),emp_vario(:,2),[0 0 0],
[],optimset('Display','off'));
  otherwise
    error('You specified a wrong variogram model');
end

%if exitflag==0
%  warning('Number of iterations exceeded options.MaxIter or number of function
evaluations exceeded options.FunEvals.');
%elseif exitflag==-1
%  warning('Algorithm was terminated by the output function.');
%elseif exitflag==-2
%  warning('Problem is infeasible: the bounds lb and ub are inconsistent.');
%elseif exitflag==-3
%  warning('Optimization could not make further progress.');
%end

a = lam(1);
b = 0;
c1 = lam(2);
c2 = lam(3);


if strcmp(detail_plots,'y')
  [y_final,dyda] = ps_variogram(model,emp_vario(:,3),a,b,c1,c2);
  figure(fig);
  plot(emp_vario(:,3),y_final,'r','linewidth',2);
end



% subfunctions

function y = ps_variogram_exp(x,xdata);

y = x(2)*(1-exp(-xdata/x(1)))+x(3);


function y = ps_variogram_gaus(x,xdata);

y = x(2)*(1-exp(-(xdata/x(1)).^2))+x(3);


function y = ps_variogram_spher(x,xdata);

y = x(2)*(1.5*min(xdata/x(1),1)-0.5*min(xdata/x(1),1).^3)+x(3);
```